



Complex Social &
Computational Systems

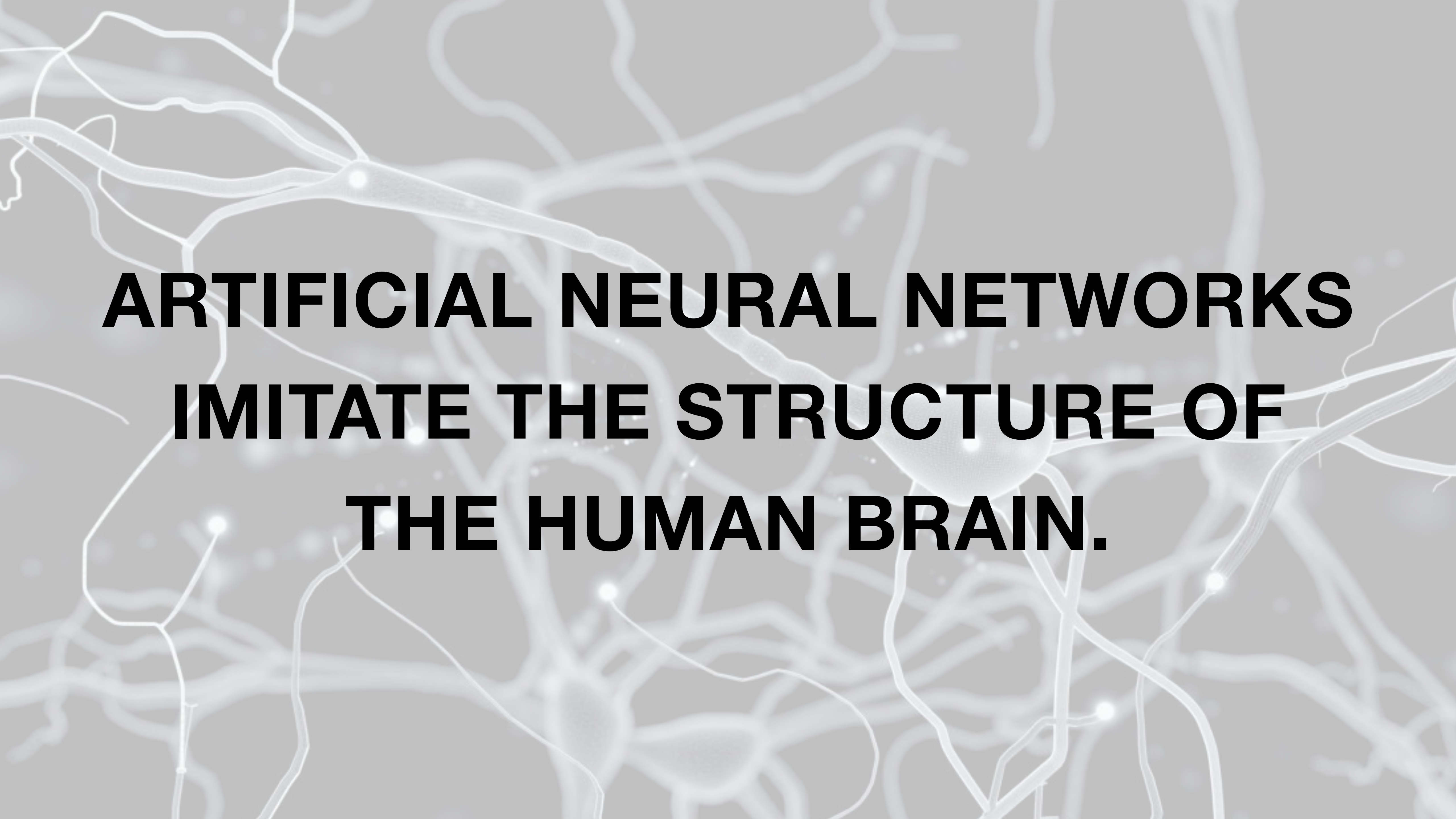
NEURAL NETWORKS I

— THE BASICS

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING
ZURICH, MARCH 16-20, 2026

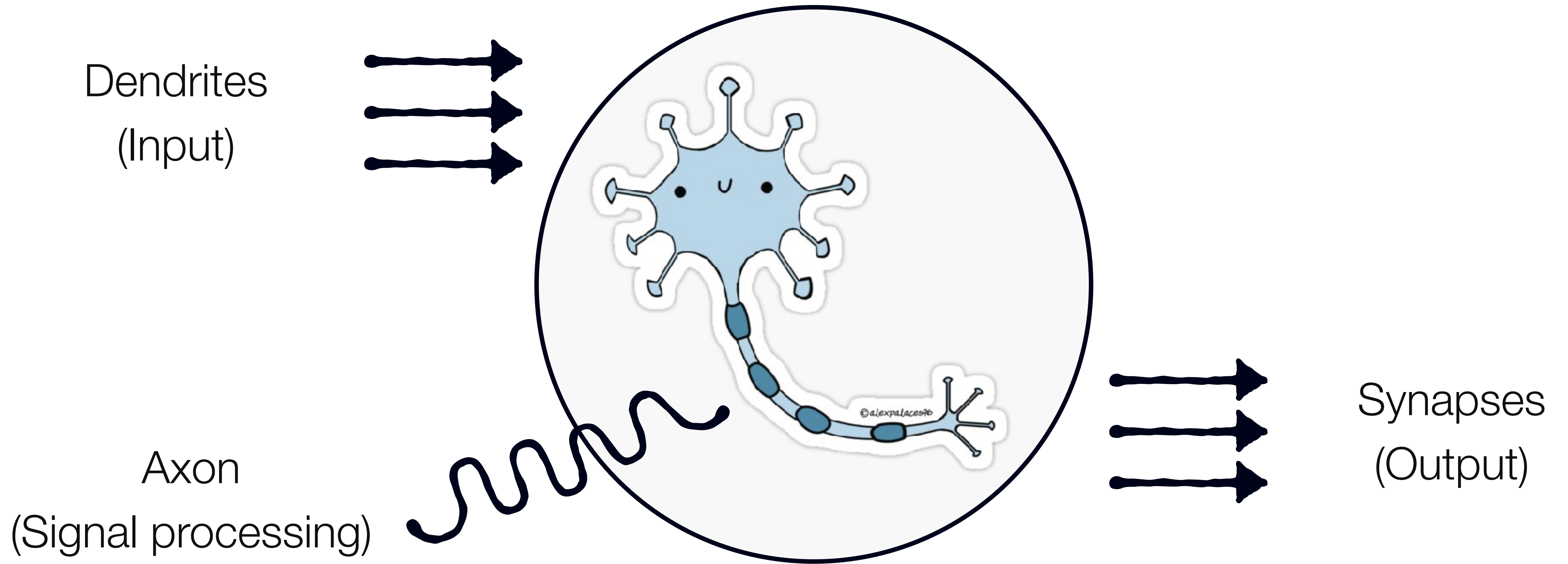
**HUMAN
INTELLIGENCE
IS STILL
UNPARALLELED.**



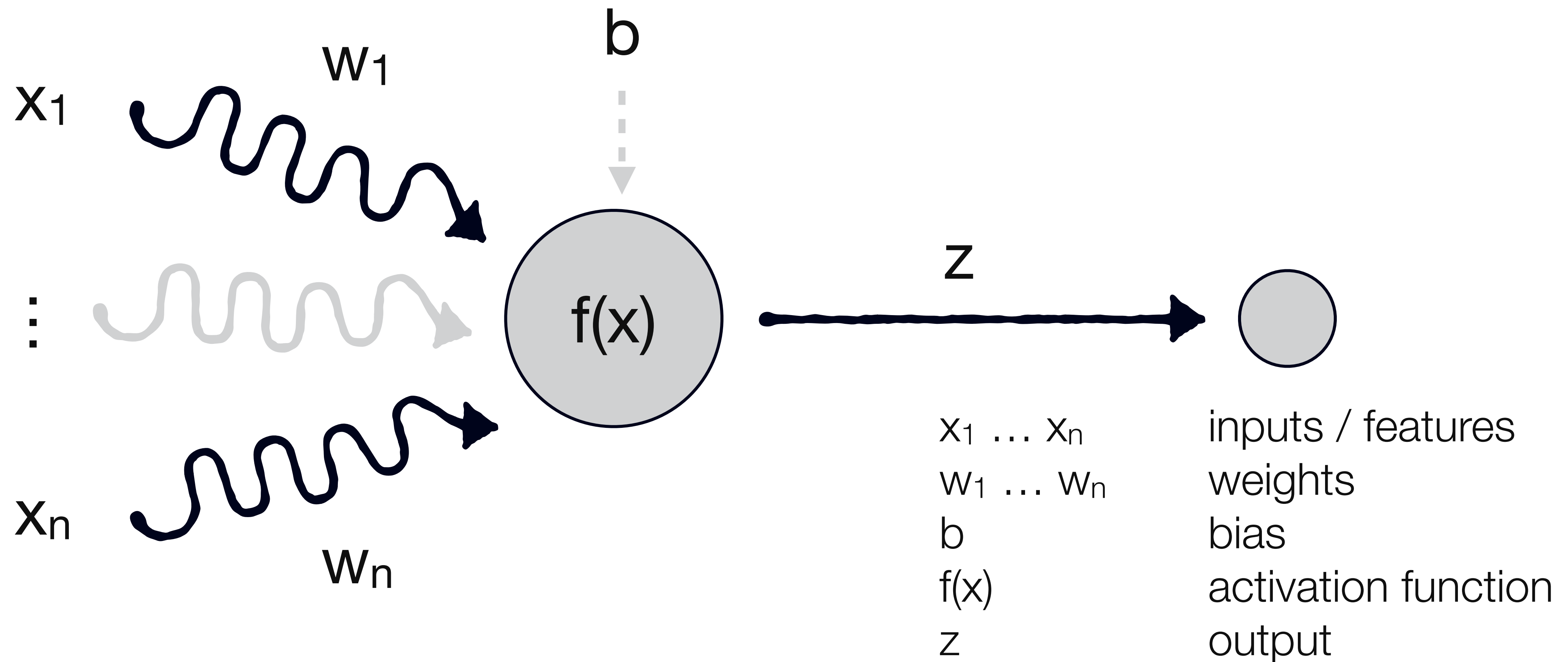


**ARTIFICIAL NEURAL NETWORKS
IMITATE THE STRUCTURE OF
THE HUMAN BRAIN.**

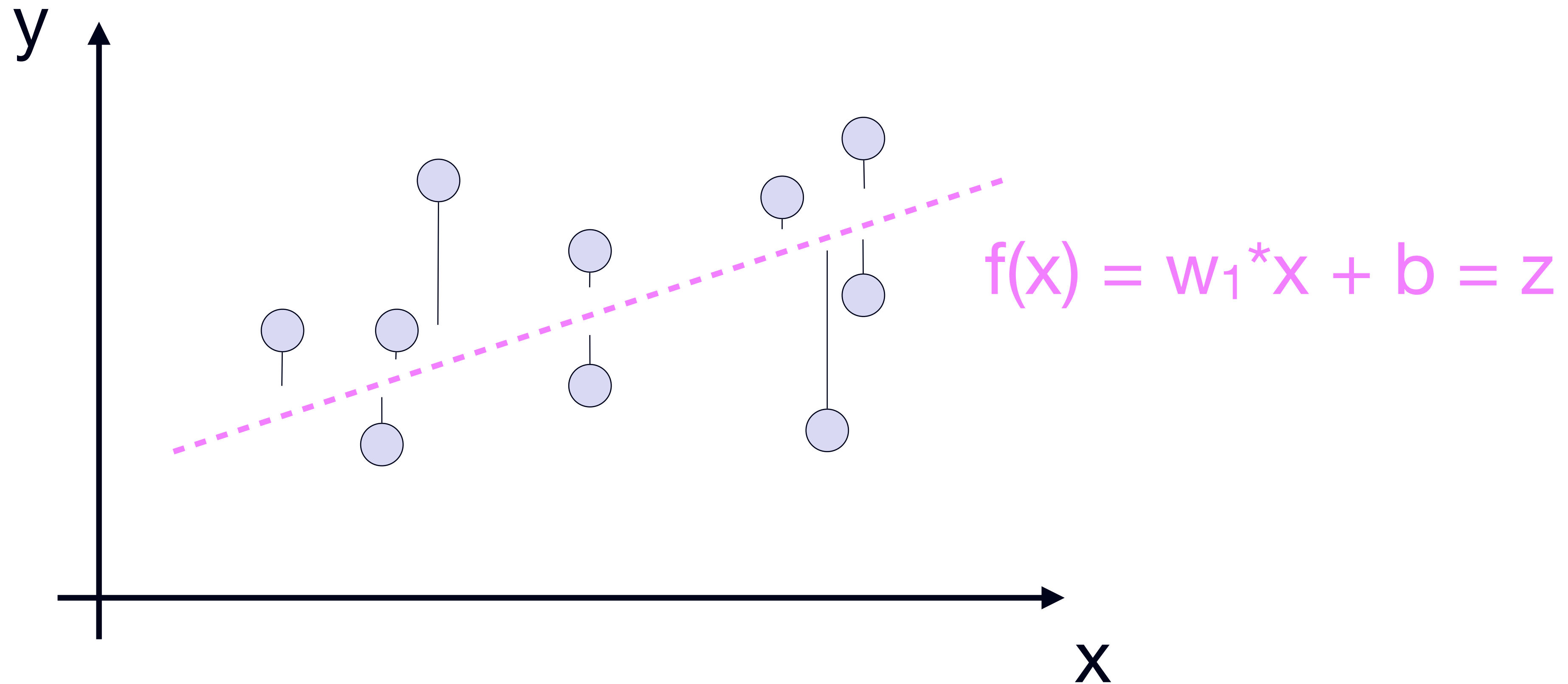
THE SIMPLEST FORM OF A NEURAL NETWORK: INSPIRED BY THE NEURON.



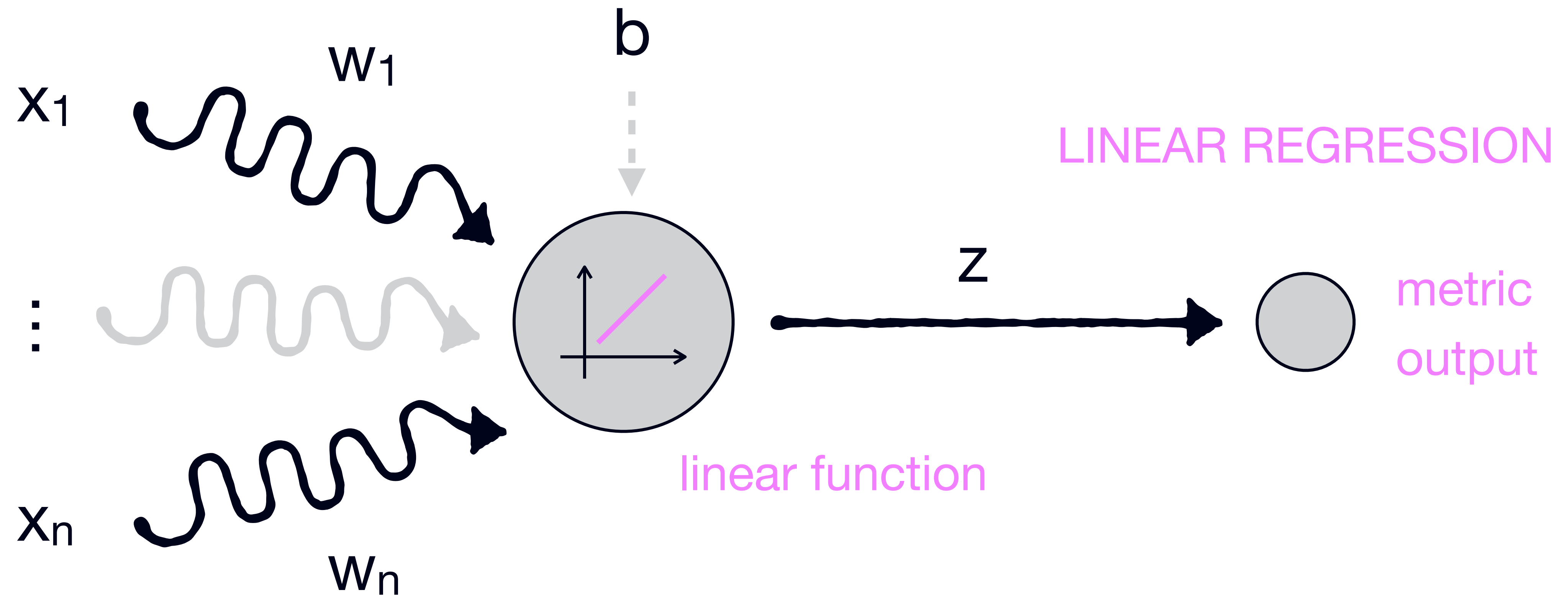
THE SIMPLEST FORM OF A NEURAL NETWORK: THE PERCEPTRON.



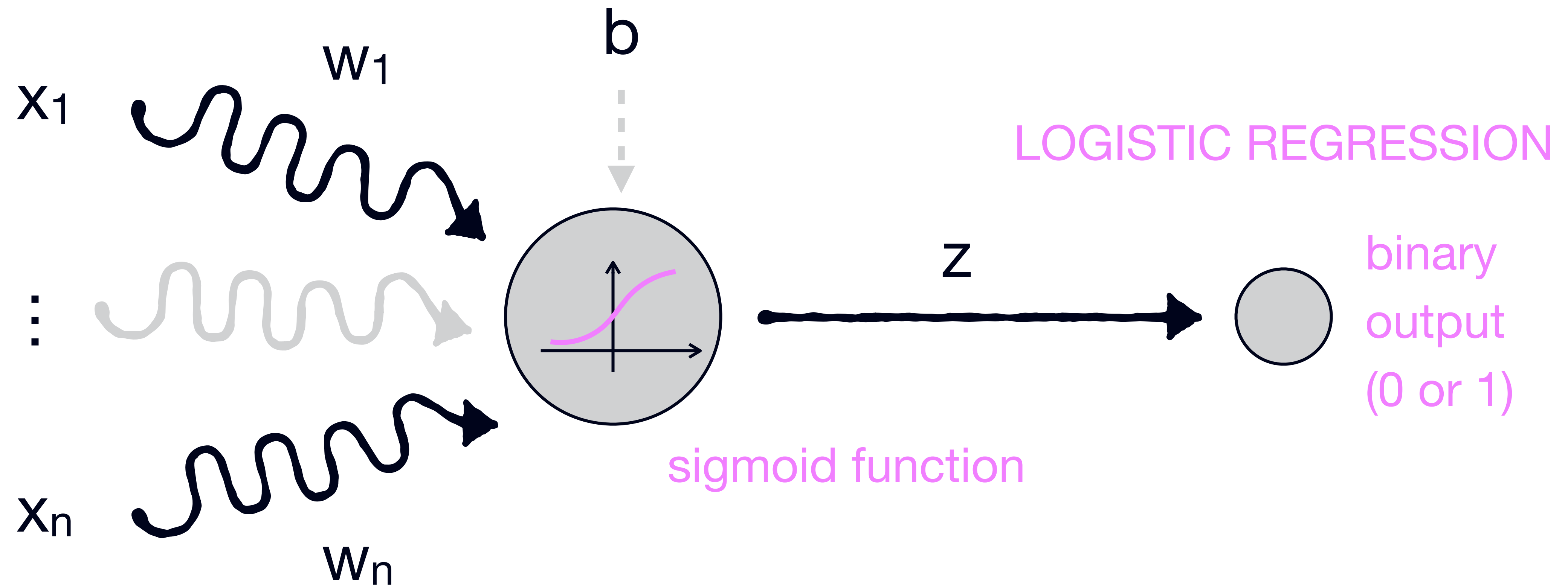
THE PERCEPTRON – A RELATIVE OF THE LINEAR REGRESSION?



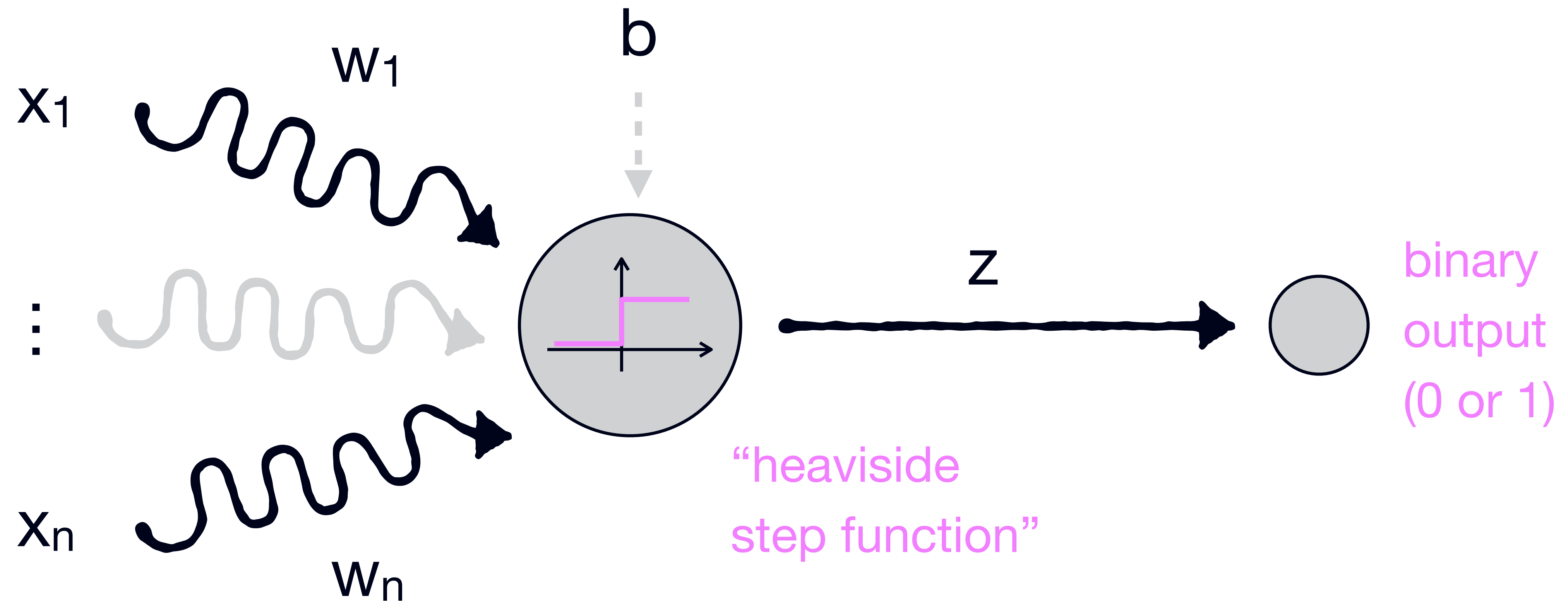
THE PERCEPTRON – A RELATIVE OF THE LINEAR REGRESSION? YES AND NO.



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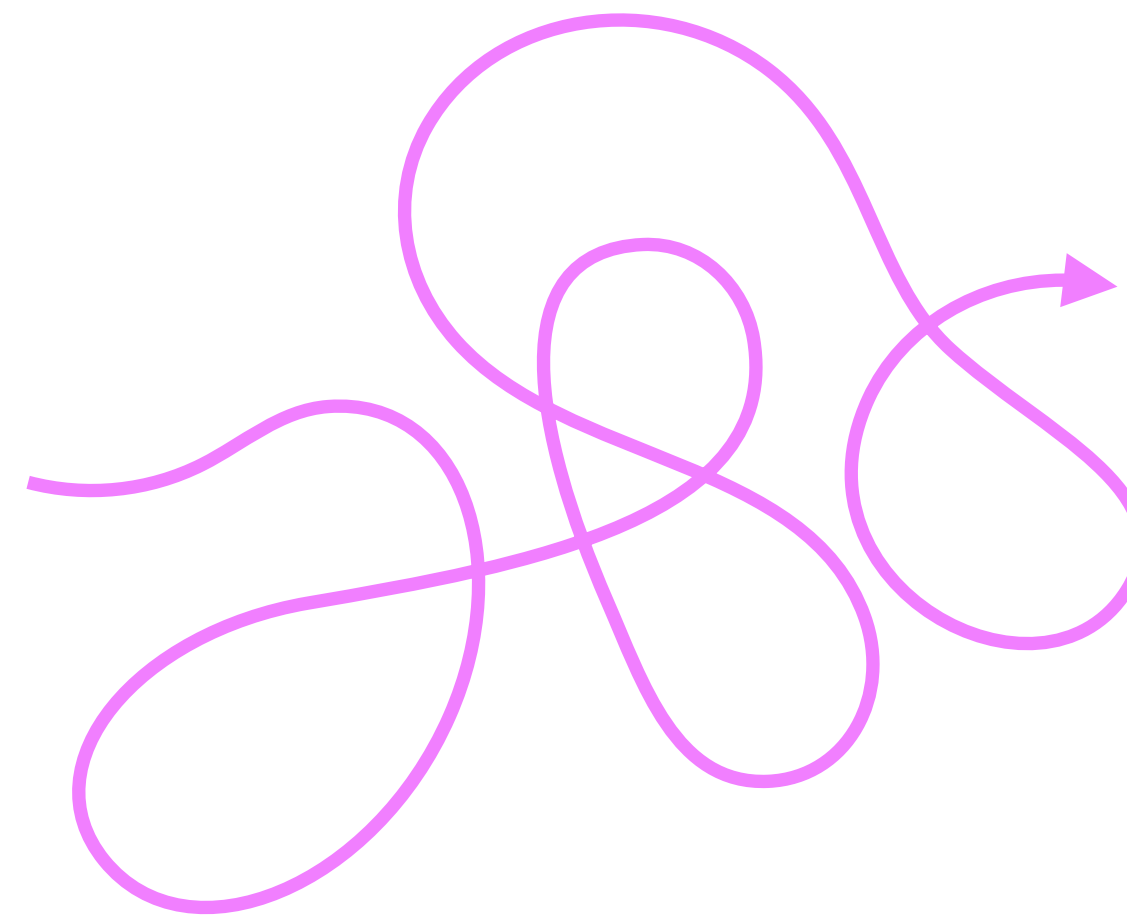
THE PERCEPTRON – A RELATIVE OF THE LINEAR REGRESSION? YES AND NO.



NOT ALL PROBLEMS CAN BE SOLVED WITH A SIMPLE REGRESSION MODEL.

Which individuals develop borderline personality disorder in the presence of which risk and protective factors?

Experiences of abuse
Emotional skills A, B, C, ...
Genetic factors A, B, C, ...
Social factors A, B, C, ...
...

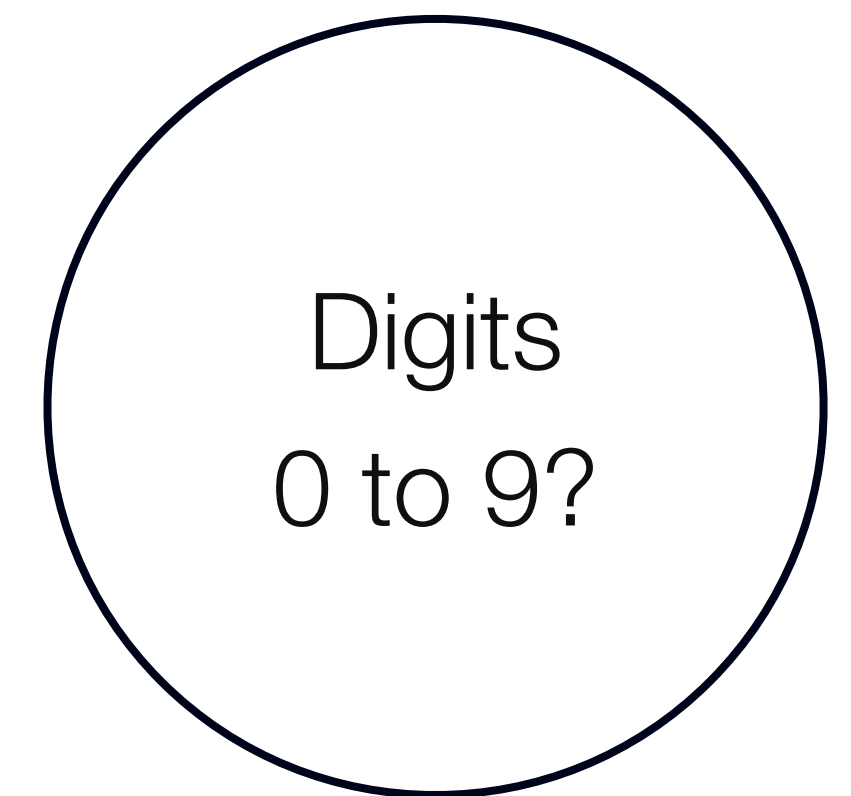
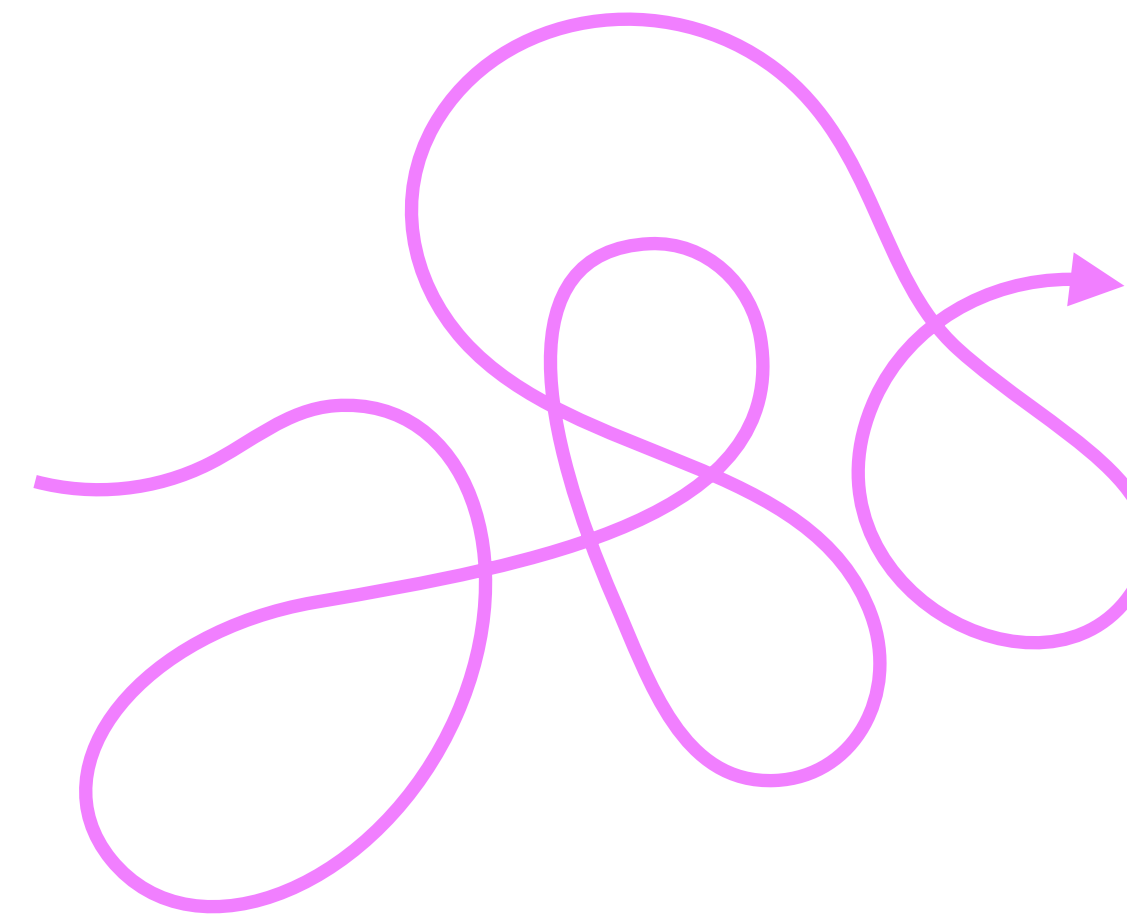
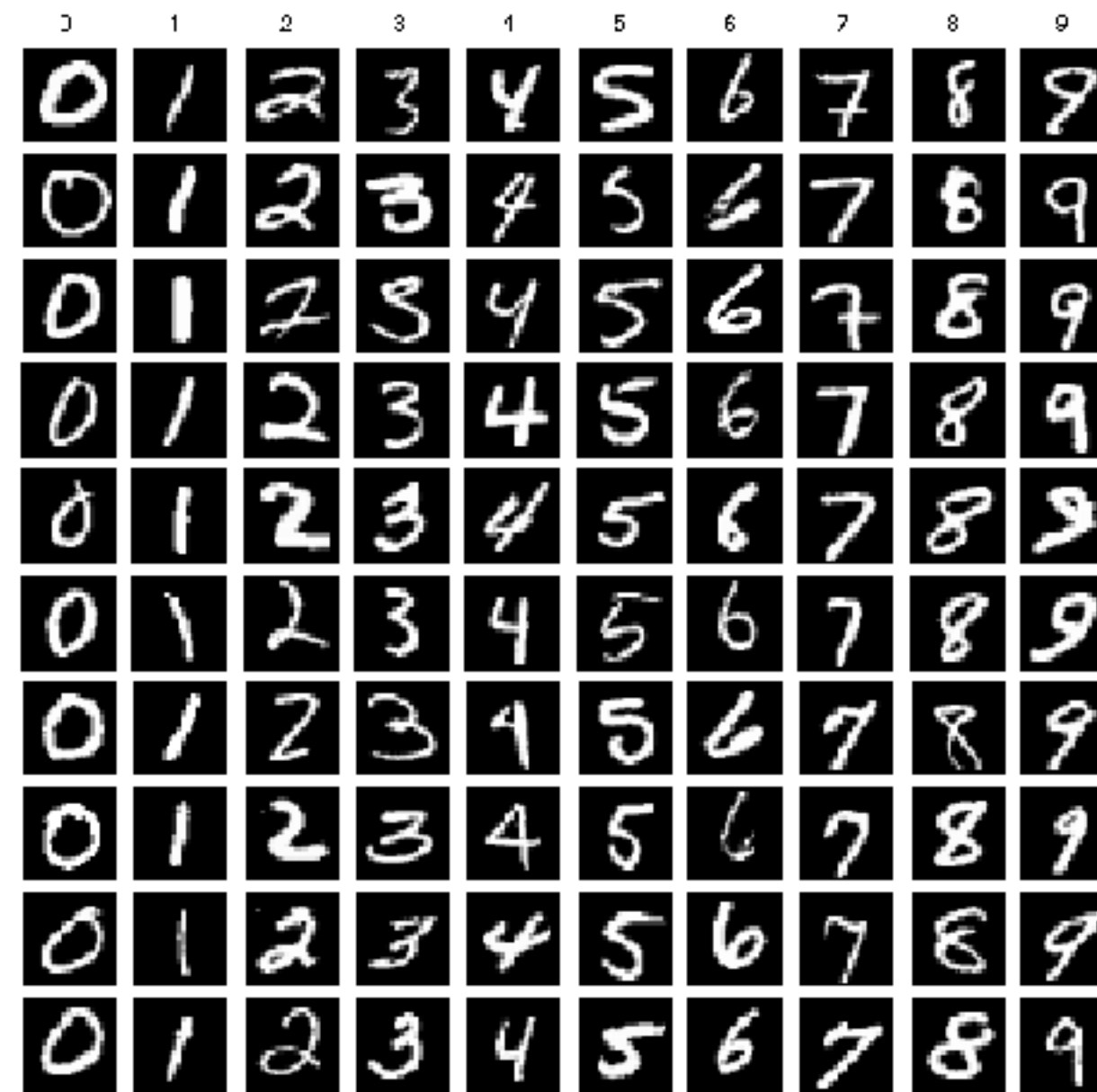


Borderline
yes/no?

NOT ALL PROBLEMS CAN BE SOLVED WITH A SIMPLE REGRESSION MODEL.

A classic machine learning example: Recognizing handwritten digits.

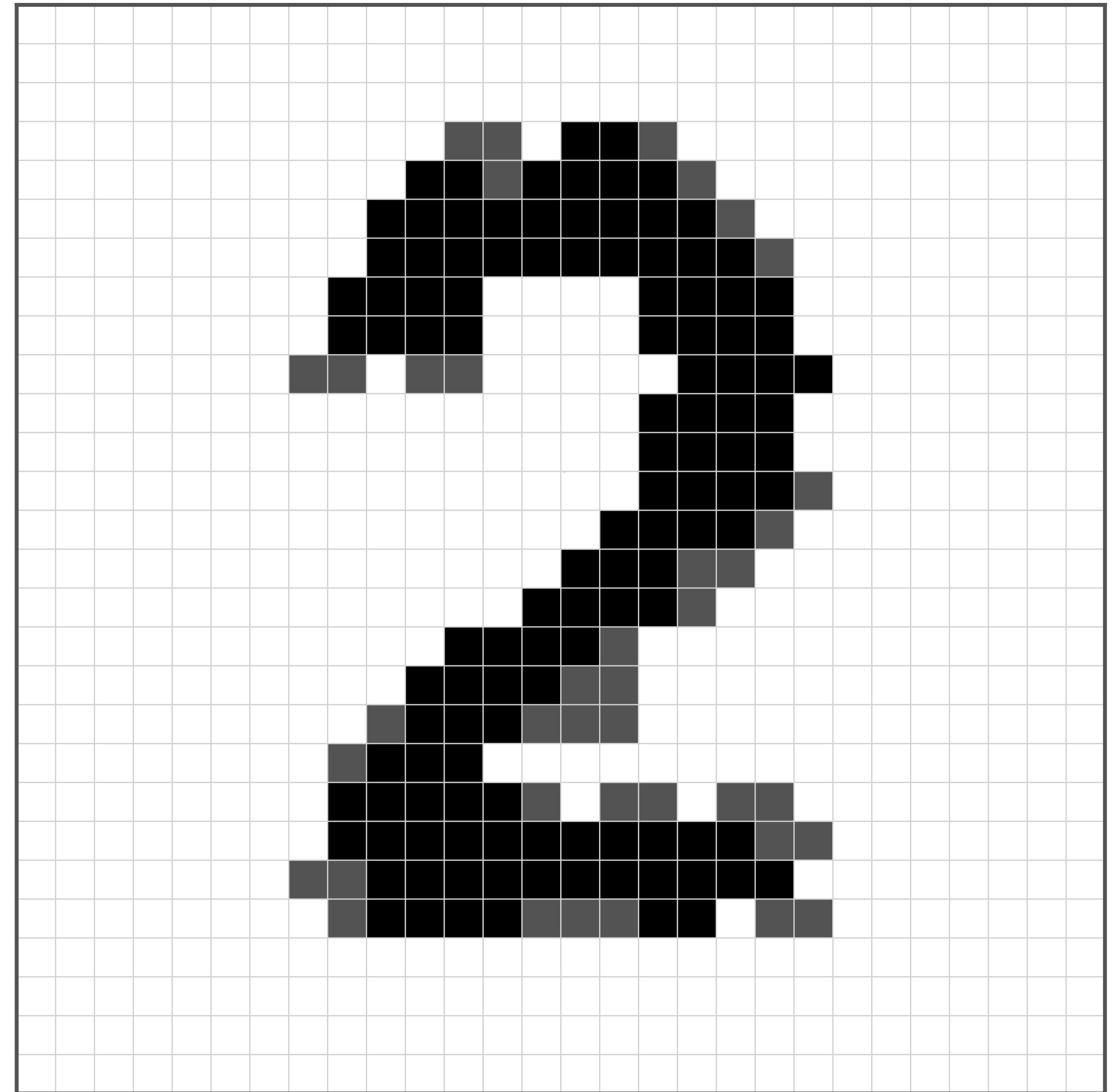
MNIST dataset
60,000 pictures with labels
28x28 pixels
in grayscale



AN MNIST EXAMPLE

28 x 28

= 784 features

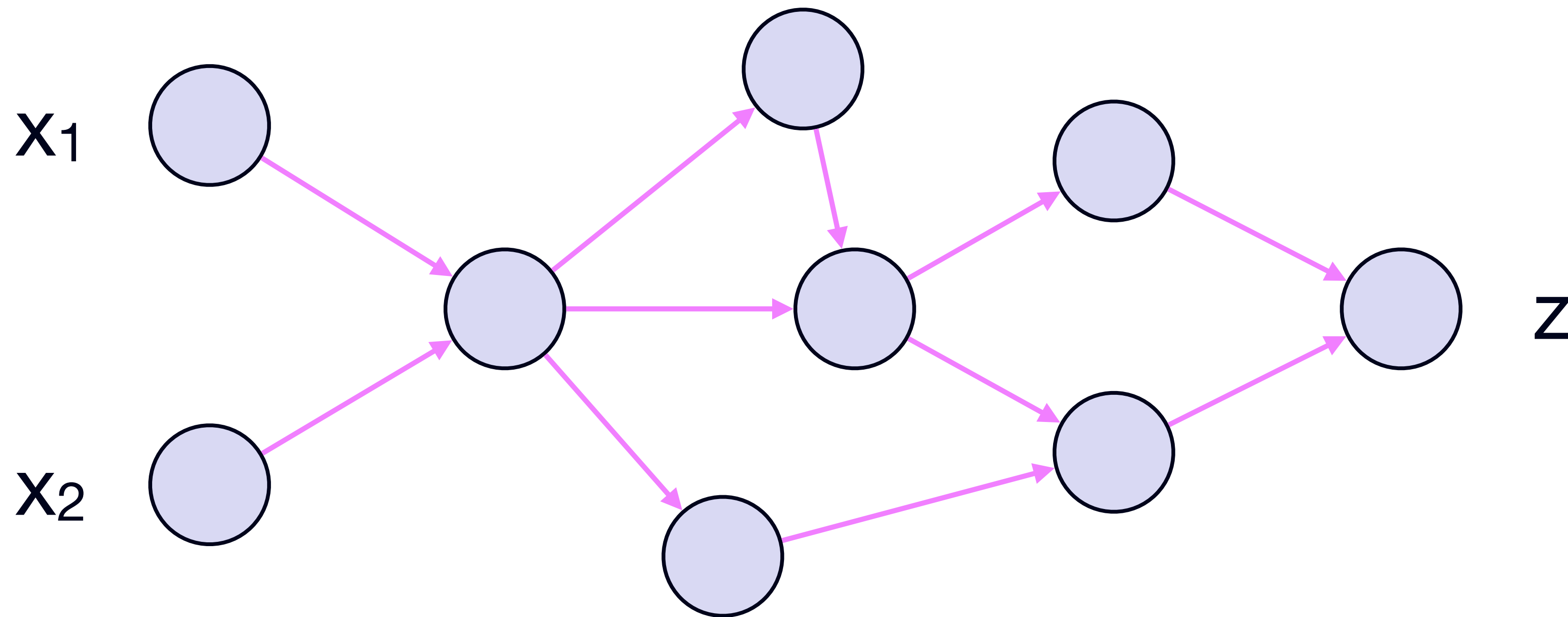


**A SOLUTION FOR COMPLEX
PROBLEMS: STACK IT UP!**

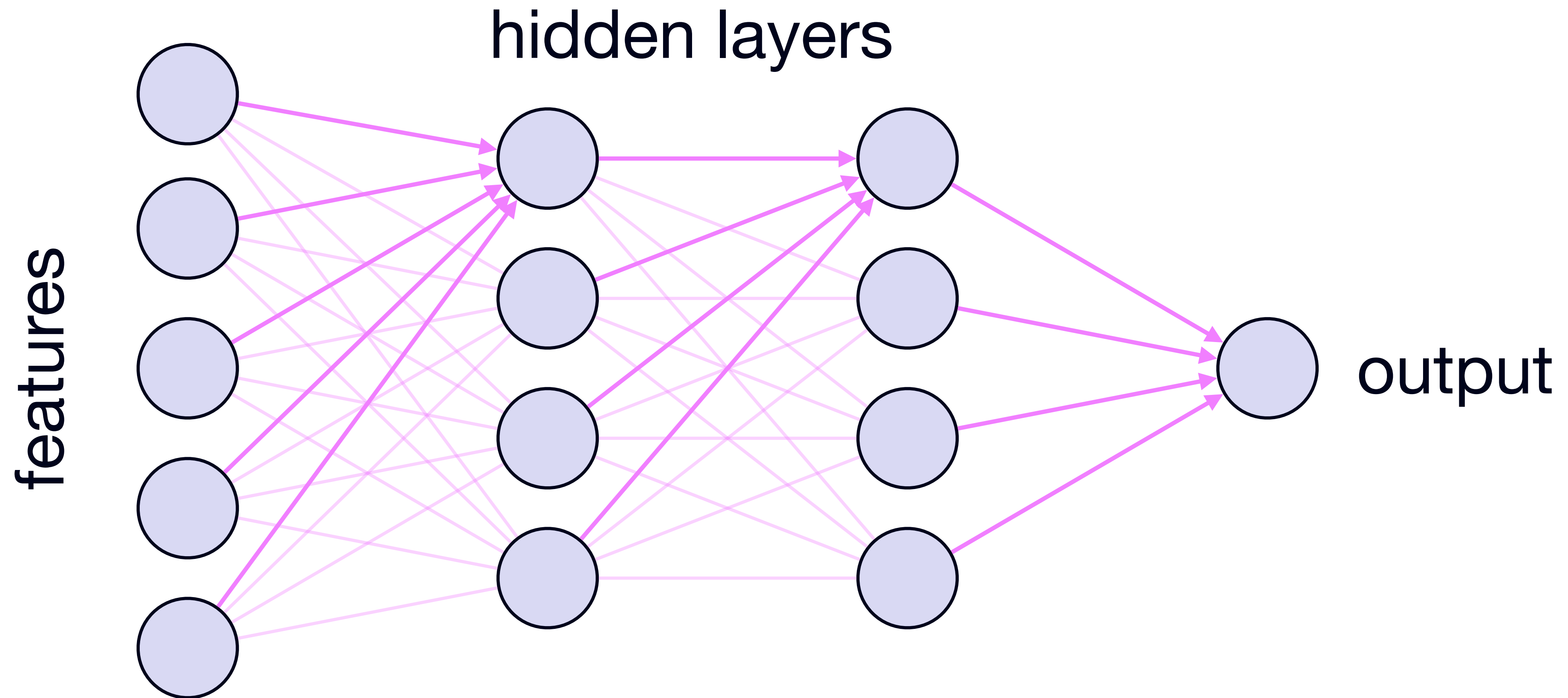


CONSTRUCTING A FEED FORWARD NETWORK FROM THE PERCEPTRON.

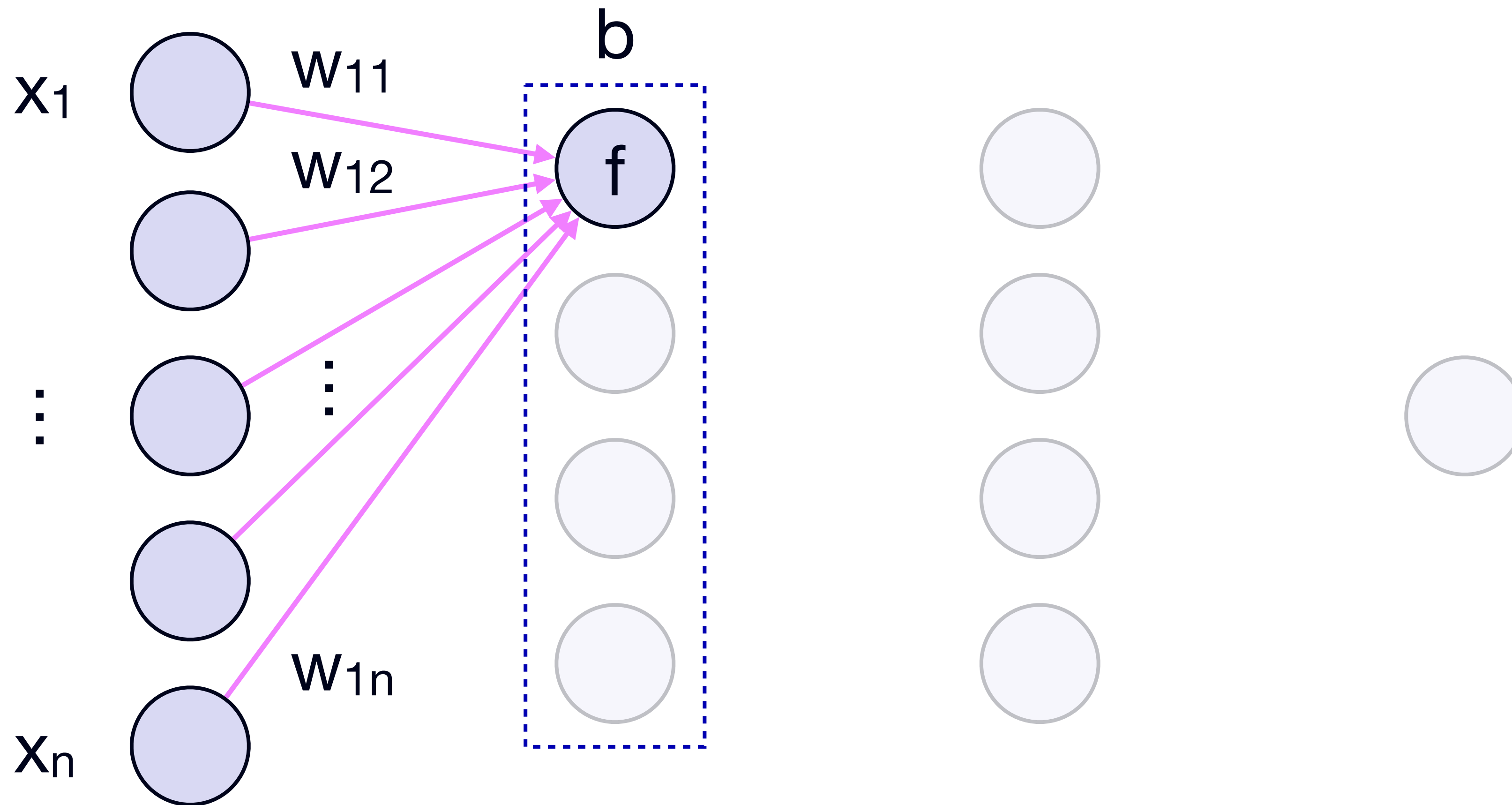
A feed forward network is also called a multi-layer perceptron.



A NEURAL NETWORK HAS AN INPUT AND OUTPUT LAYER, AND HIDDEN LAYERS.



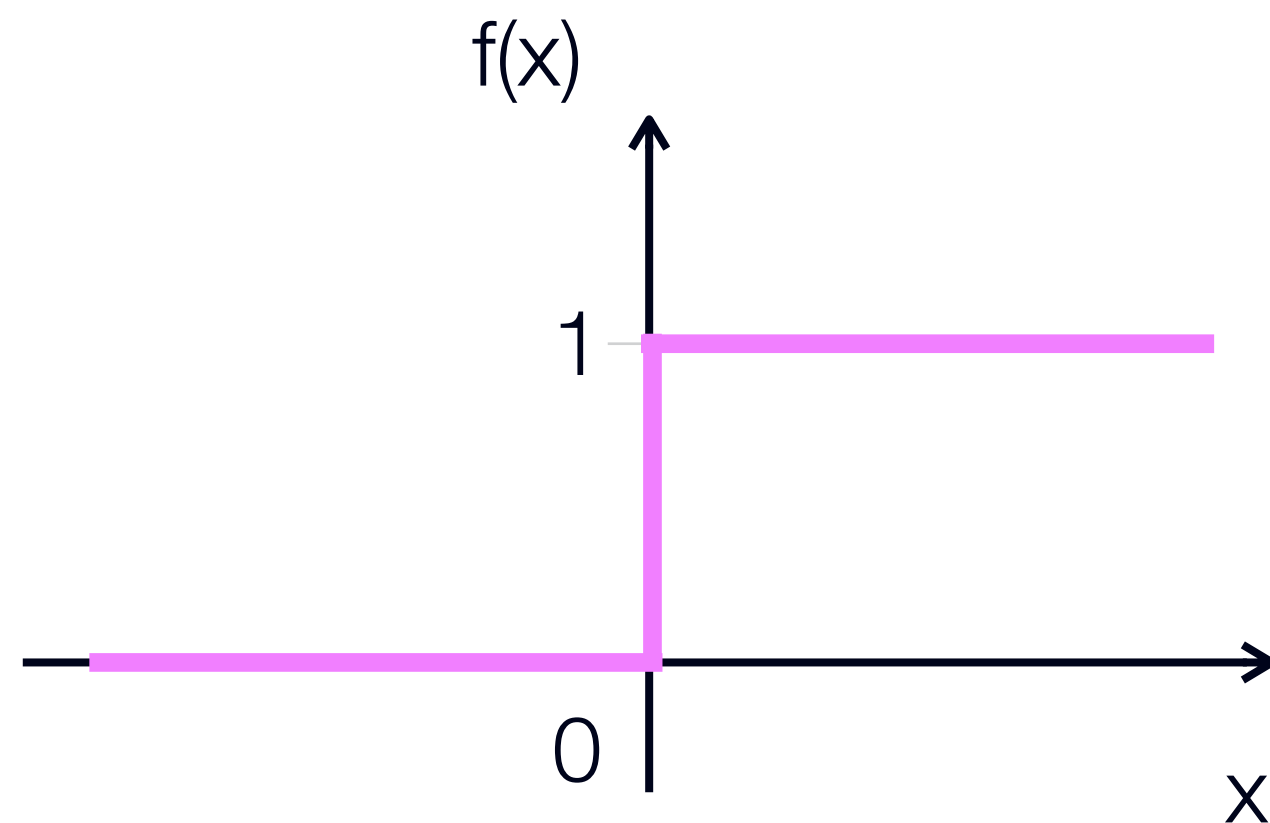
THE WEIGHTS DEFINE THE RELATIONSHIP STRENGTH BETWEEN NEURONS.



THE MOST PROMINENT ACTIVATION FUNCTIONS.

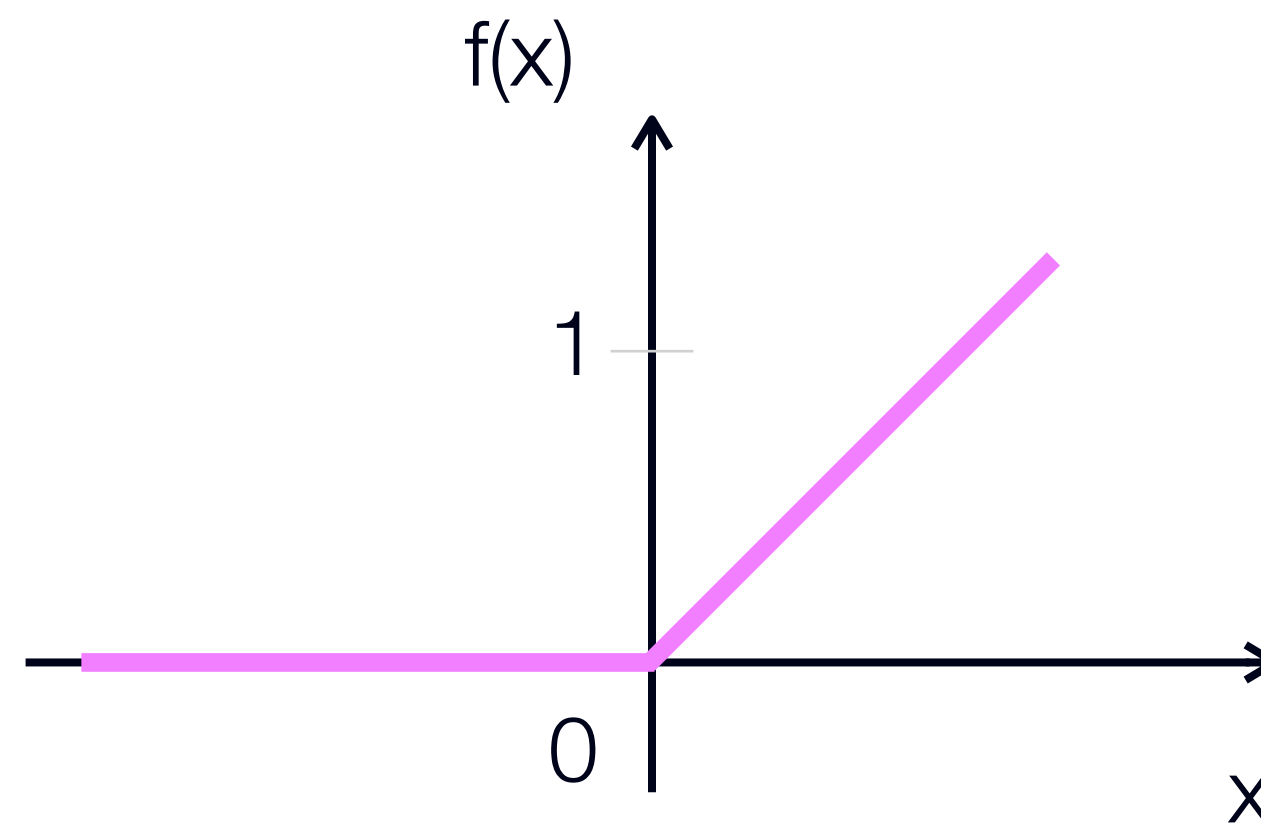
Heaviside Step Function

$$x \mapsto \begin{cases} 0 : & x < 0 \\ 1 : & x \geq 0 \end{cases}$$



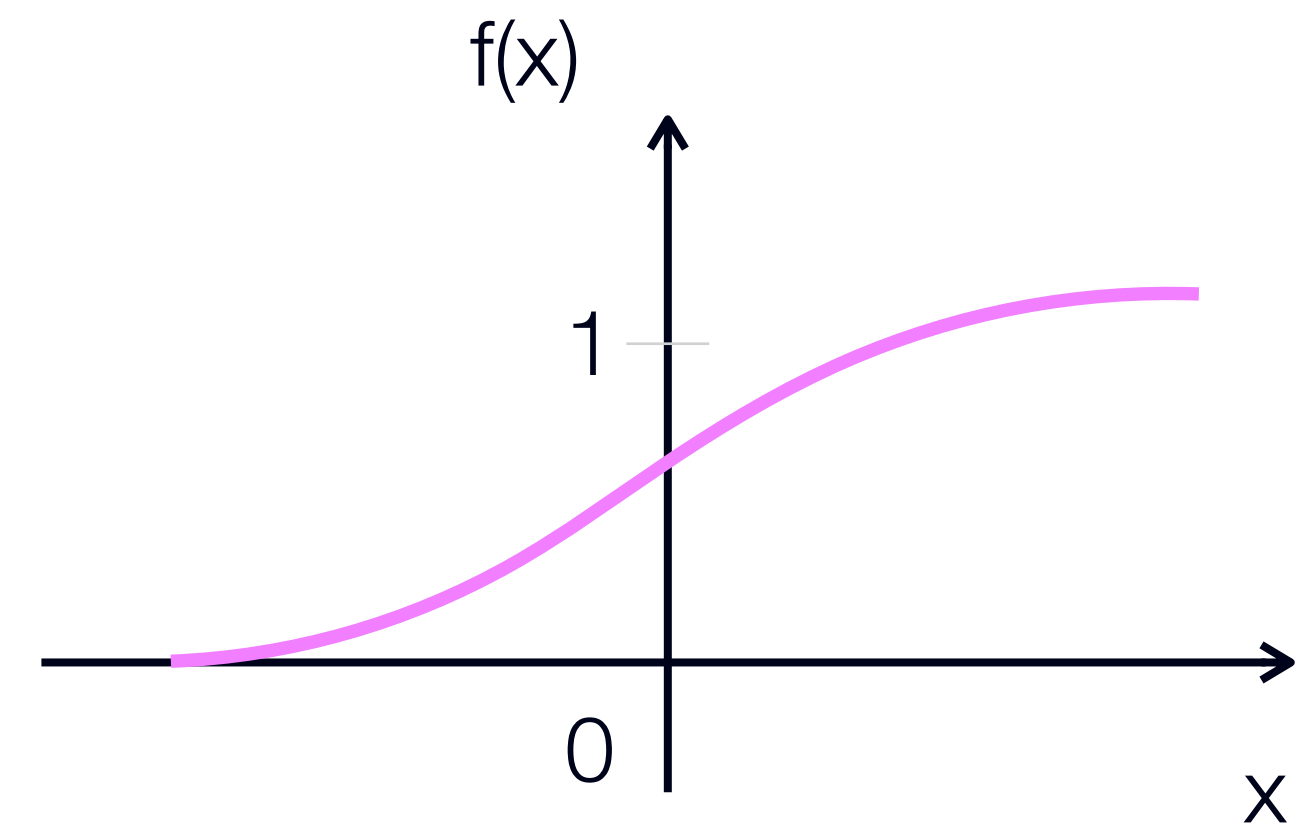
RELU Funktion

$$x \mapsto \begin{cases} 0 : & x < 0 \\ x : & x \geq 0 \end{cases}$$



Sigmoidfunktion

$$\text{sig}(t) = \frac{1}{1 + e^{-t}}$$



SOME CHARACTERISTICS OF THE NN ARE PRE-DEFINED, SOME NEED TO BE TUNED.

How many neurons does the input layer have?

How many neurons does the output layer have?

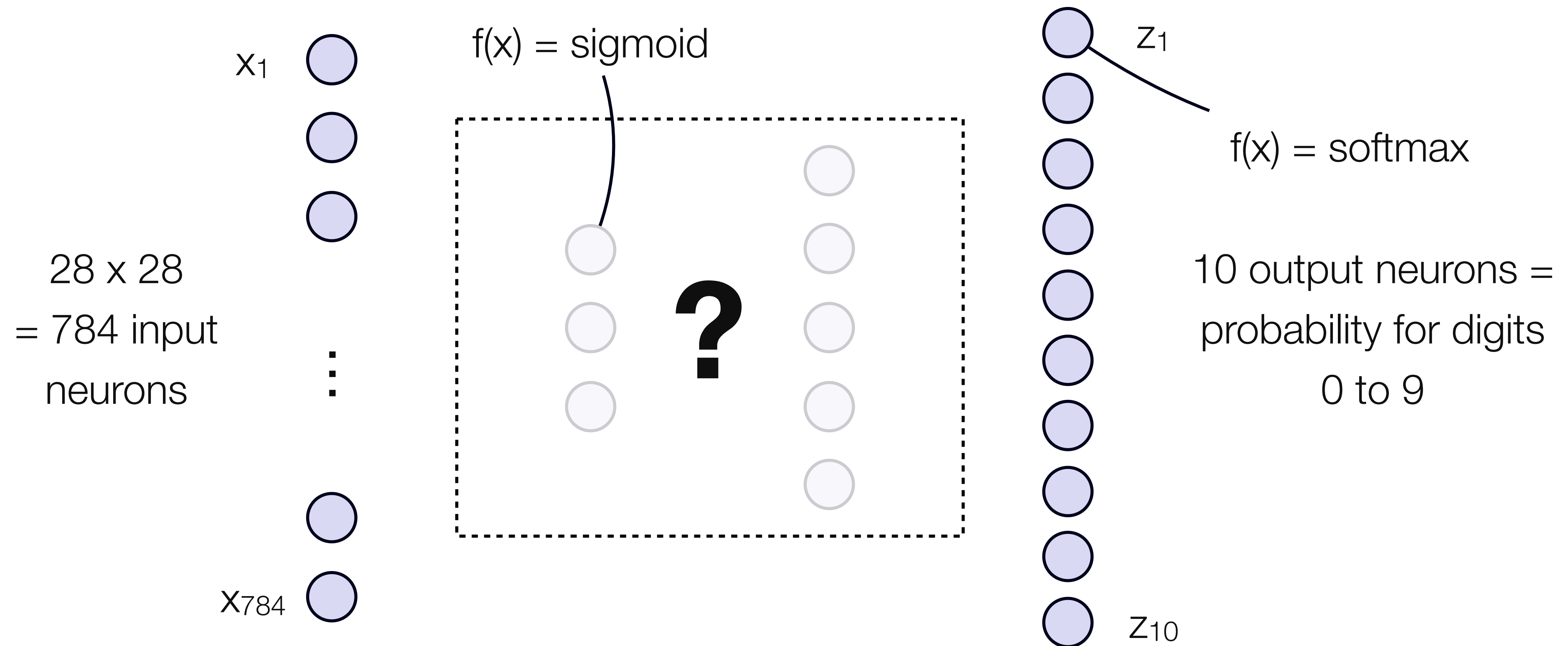
How many hidden layers are there?

How many neurons does a hidden layer have?

Which activation function do the output neurons have?

Which activation functions do the neurons in the hidden layers have?

EXAMPLE: A NN FOR HANDWRITTEN DIGIT RECOGNITION.



HOW ARE WE SUPPOSED TO INTERPRET ALL THESE PARAMETERS?

Even a neural network with 784 input neurons, 10 output neurons, and 2 hidden layers with 16 neurons each has around 13,000 parameters!



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xAI

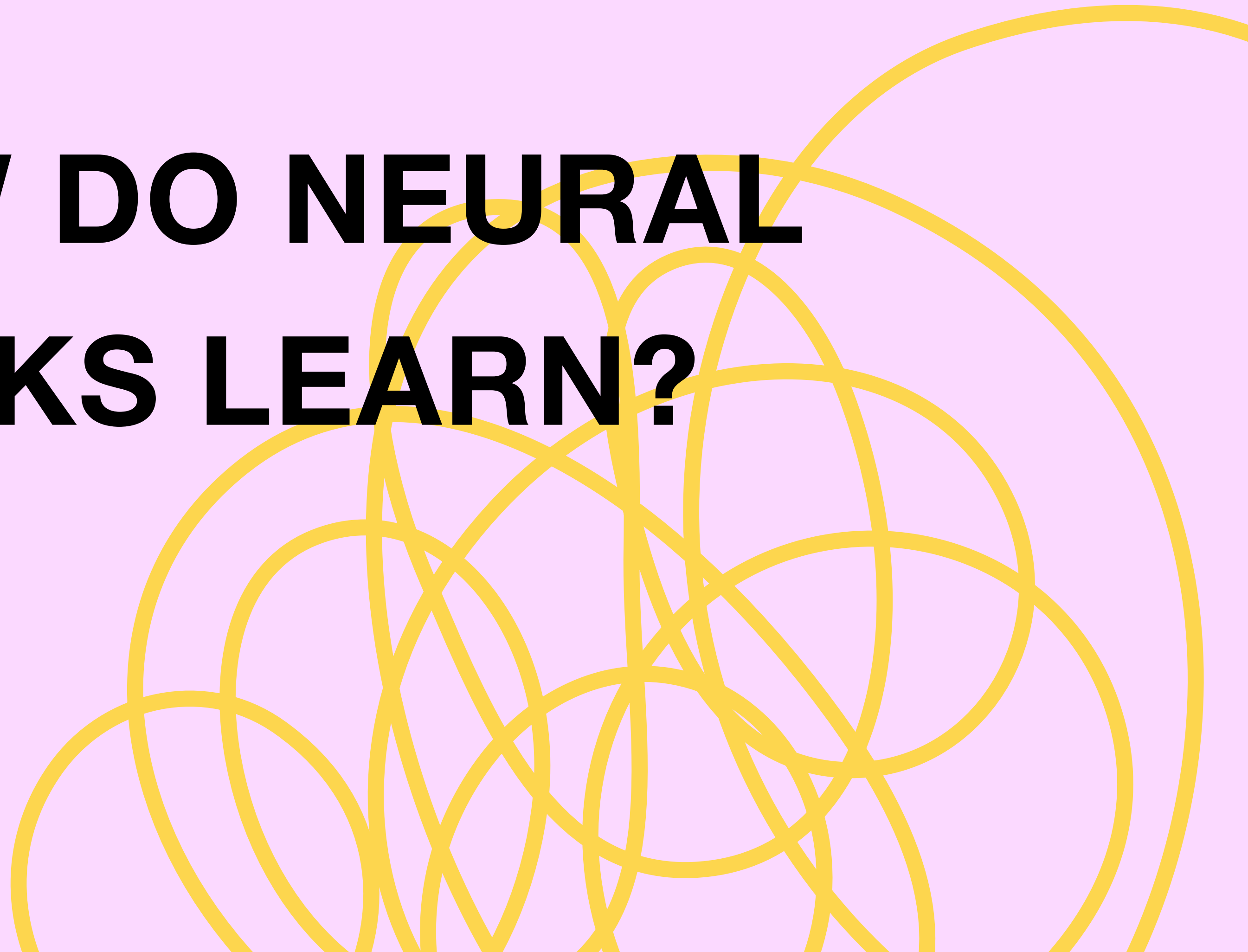
WHAT DO WE LEARN IF OUR PARAMETERS ARE NOT INTERPRETABLE?

Measuring the predictability of life outcomes with a scientific mass collaboration

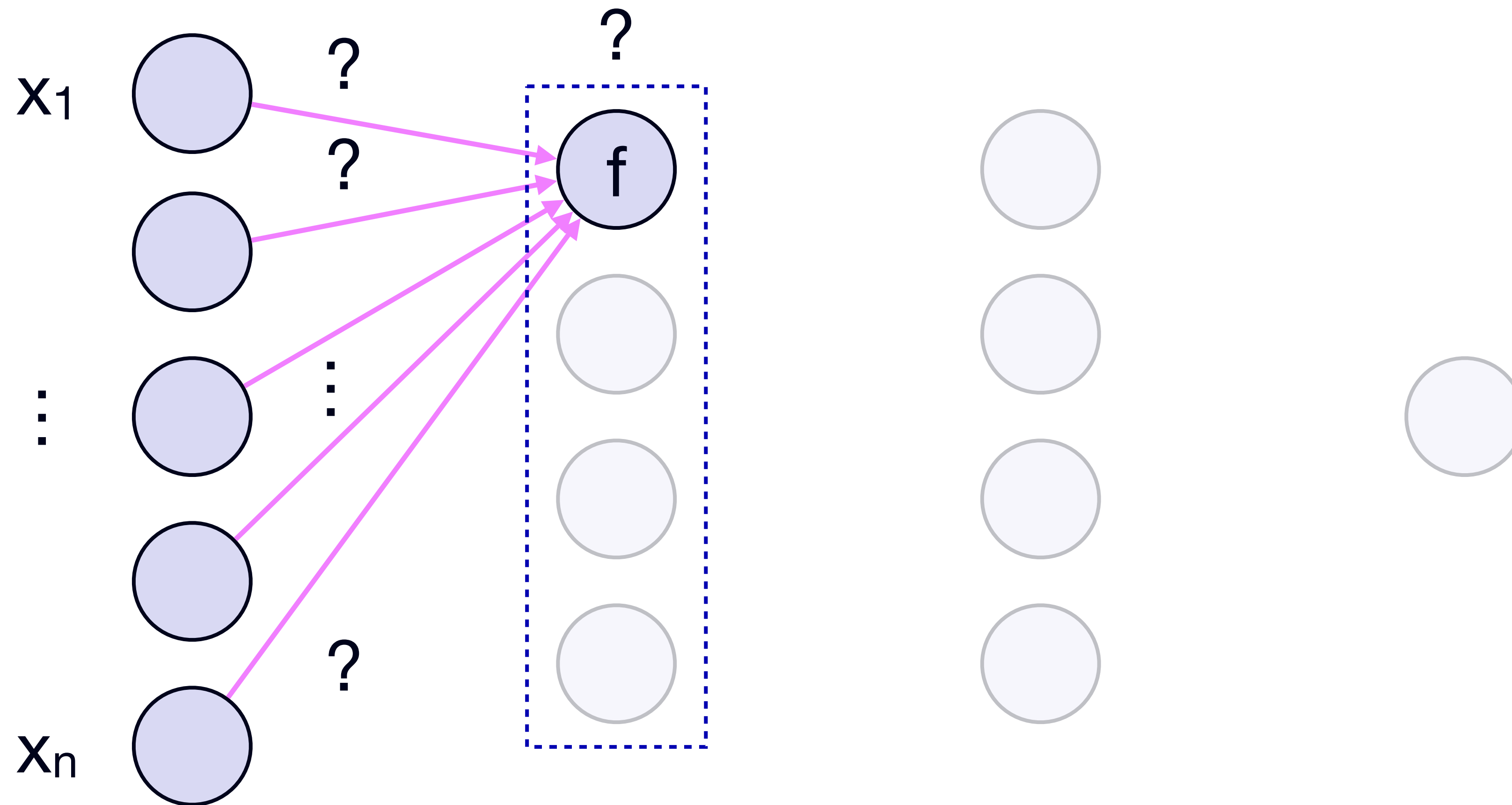
How predictable are life trajectories? We investigated this question with a scientific mass collaboration using the common task method; 160 teams built predictive models for six life outcomes using data from the Fragile Families and Child Wellbeing Study, a high-quality birth cohort study. Despite using a rich dataset and applying machine-learning methods optimized for prediction, the best predictions were not very accurate and were only slightly better than those from a simple benchmark model. Within each outcome, prediction error was strongly associated with the family being predicted and weakly associated with the technique used to generate the prediction. Overall, these results suggest practical limits to the predictability of life outcomes in some settings and illustrate the value of mass collaborations in the social sciences.

life course | prediction | machine learning | mass collaboration

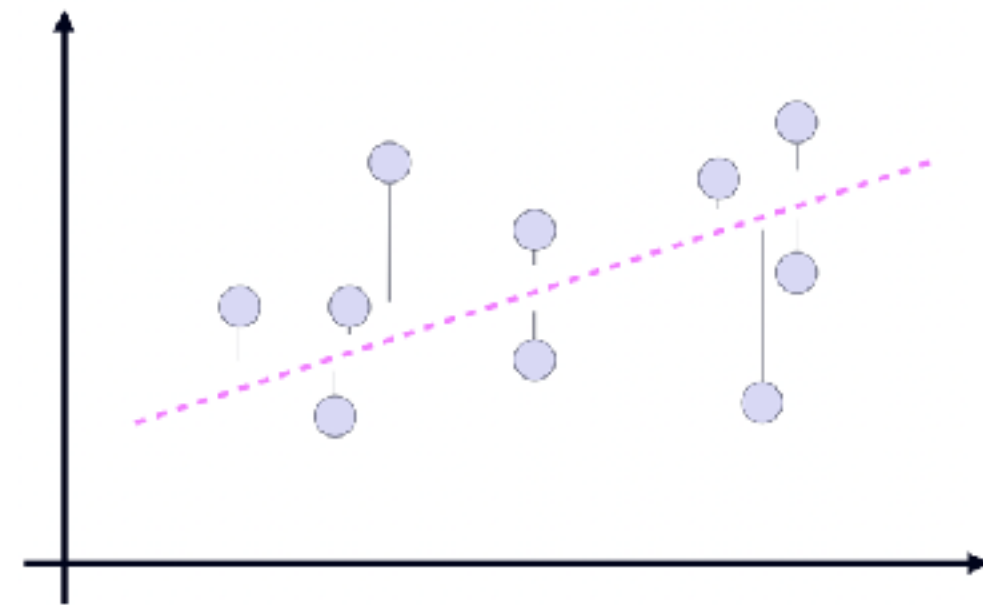
**THEN HOW DO NEURAL
NETWORKS LEARN?**



STEP ONE: INITIALIZING THE PARAMETERS RANDOMLY.



STEP TWO: MEASURING MODEL FIT.



$$\Delta(\text{blue circle}, \text{pink circle})^2$$

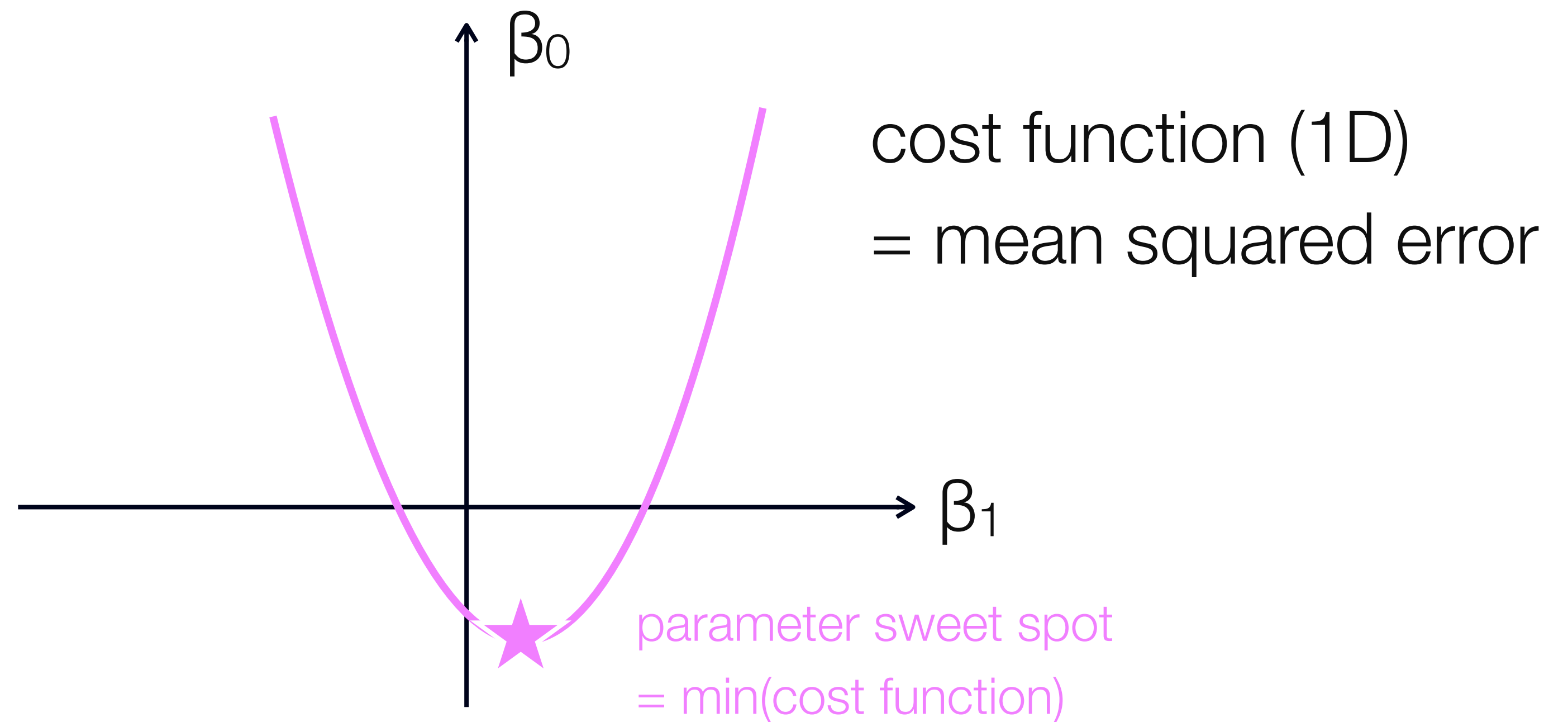
$$\Delta(\text{blue circle}, \text{pink circle})^2$$

⋮

$$\Delta(\text{blue circle}, \text{pink circle})^2$$

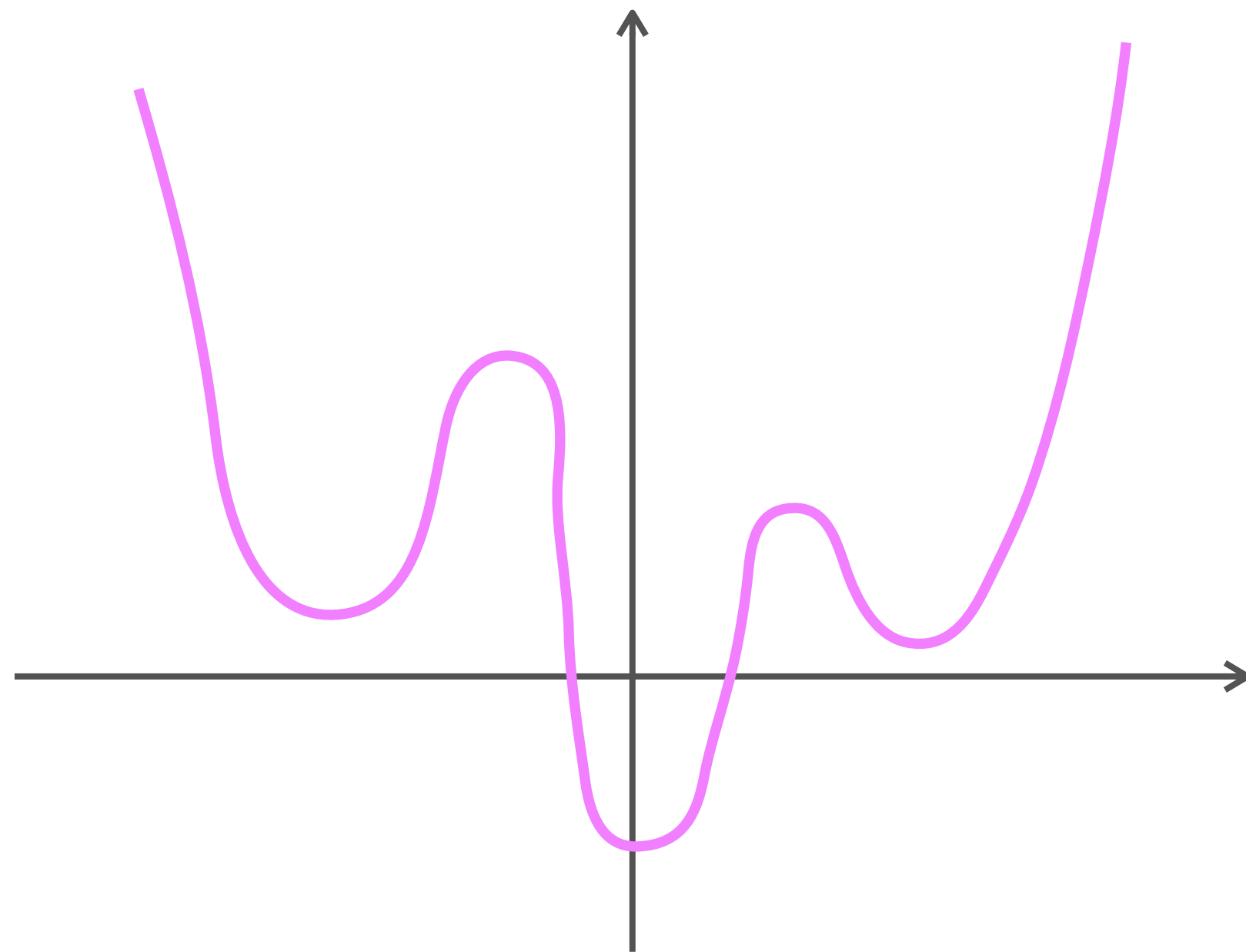
$$\frac{\Sigma}{N}$$

Just like in linear regression, we measure the mean squared error.

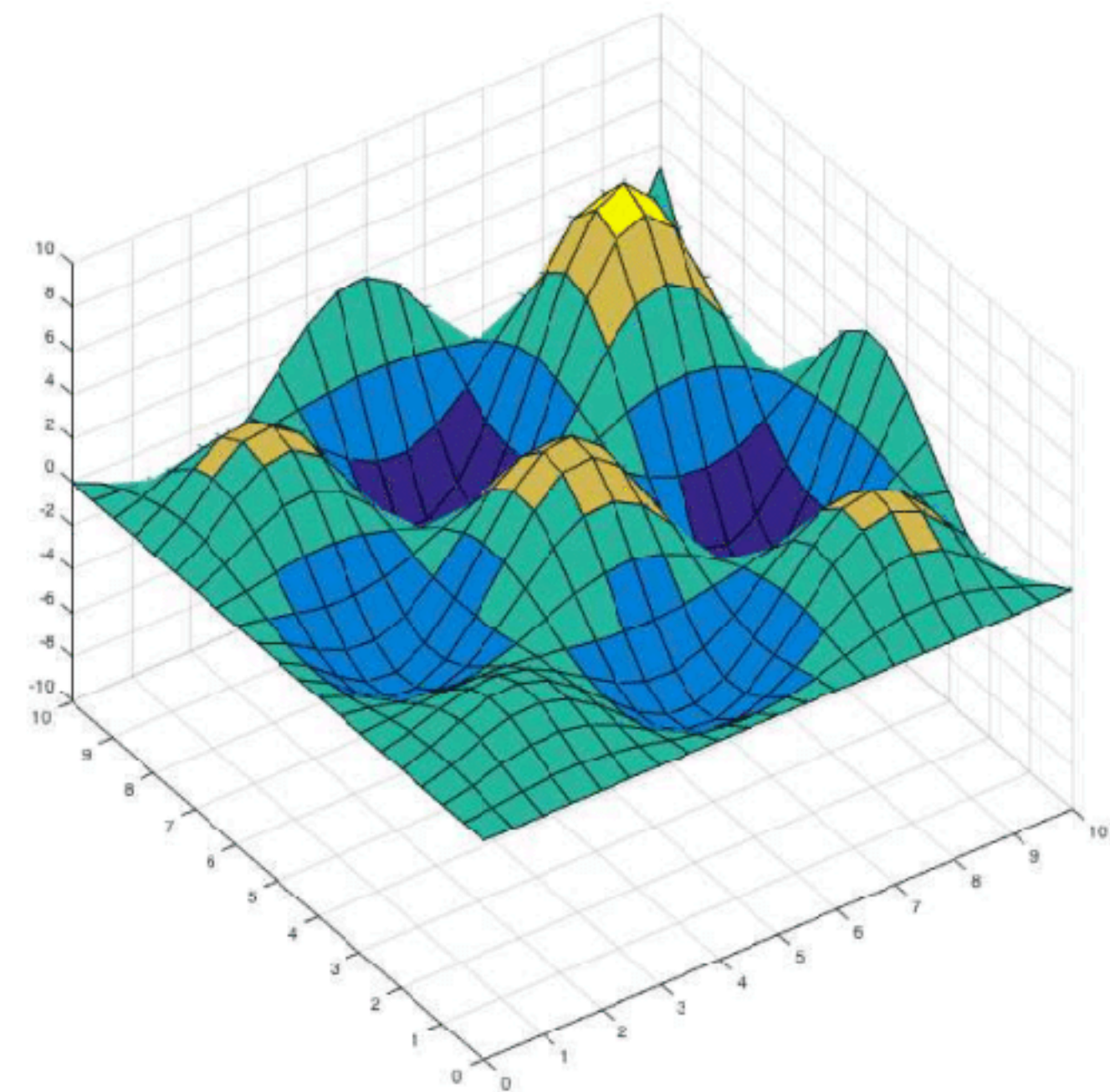


WITH NEURAL NETWORKS IT IS A BIT MORE COMPLICATED.

The cost function is non-convex.

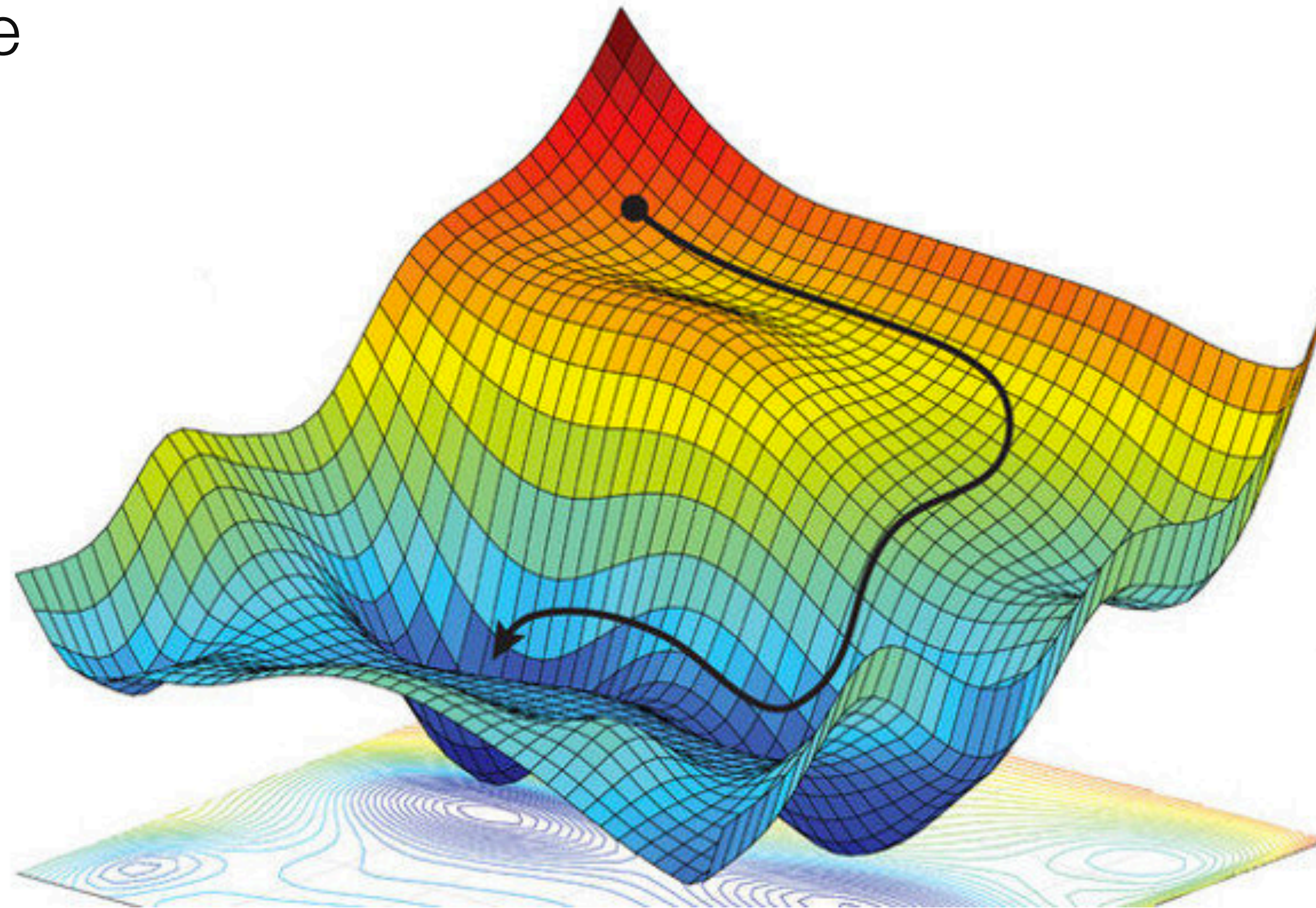


Dimensionality is high.



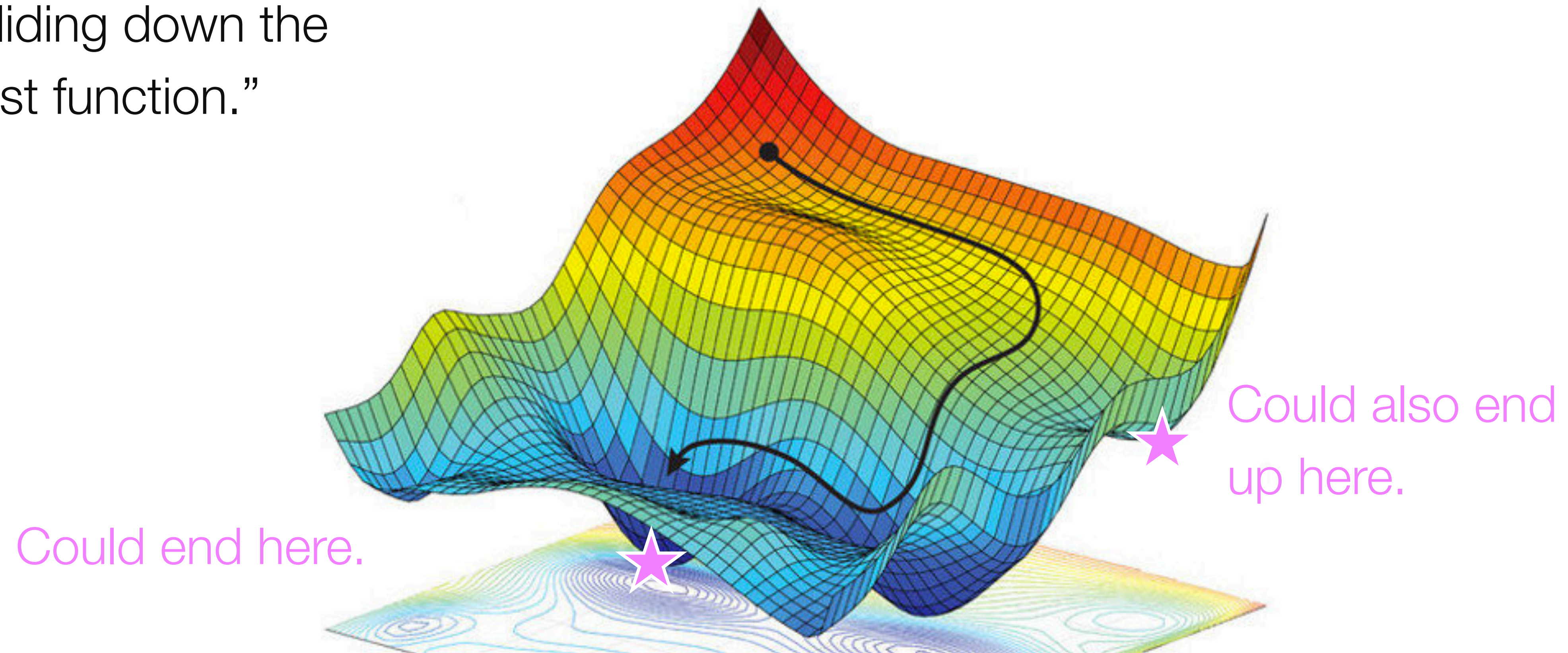
THE SOLUTION: GRADIENT DESCENT.

“Sliding down the
cost function.”



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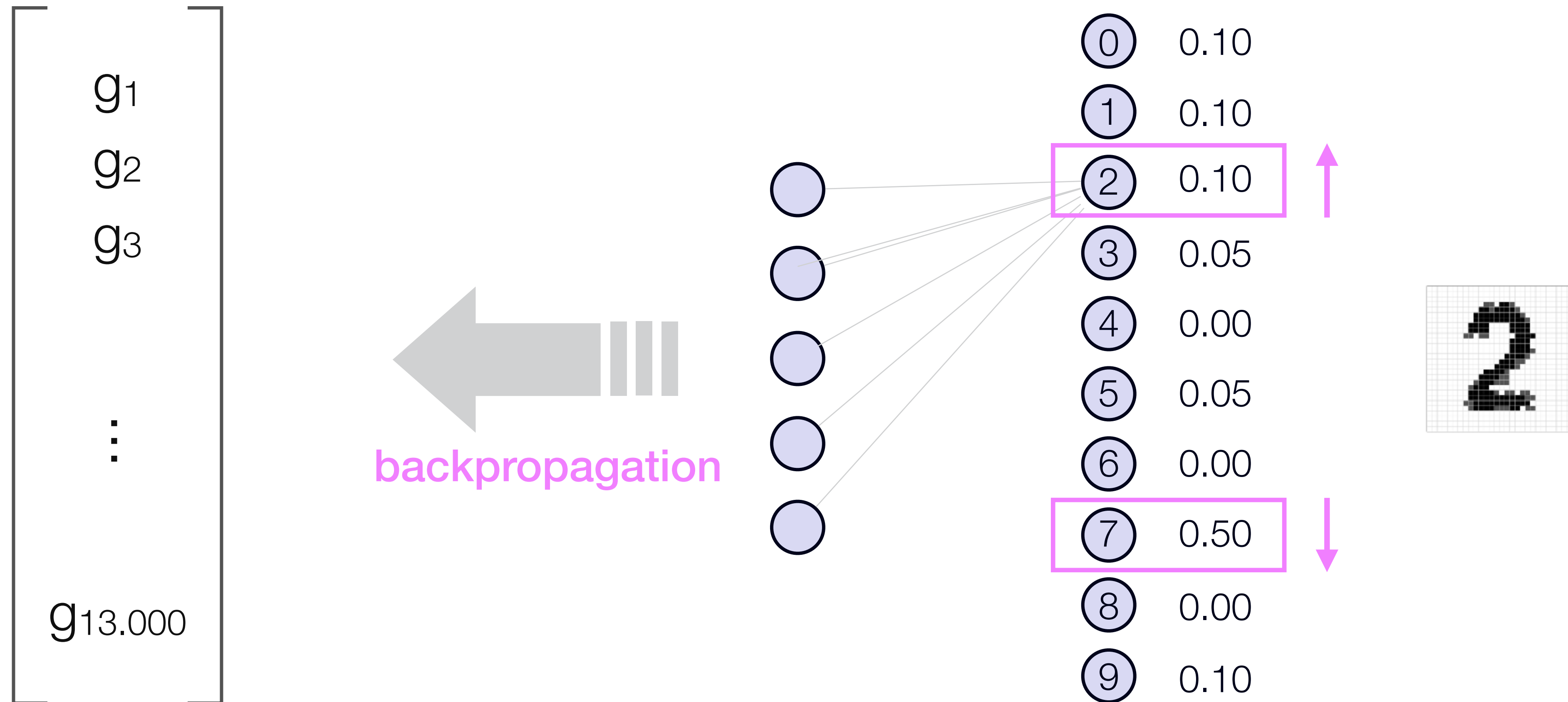


STEP THREE: BACKPROPAGATION OR UPDATING THE PARAMETERS.

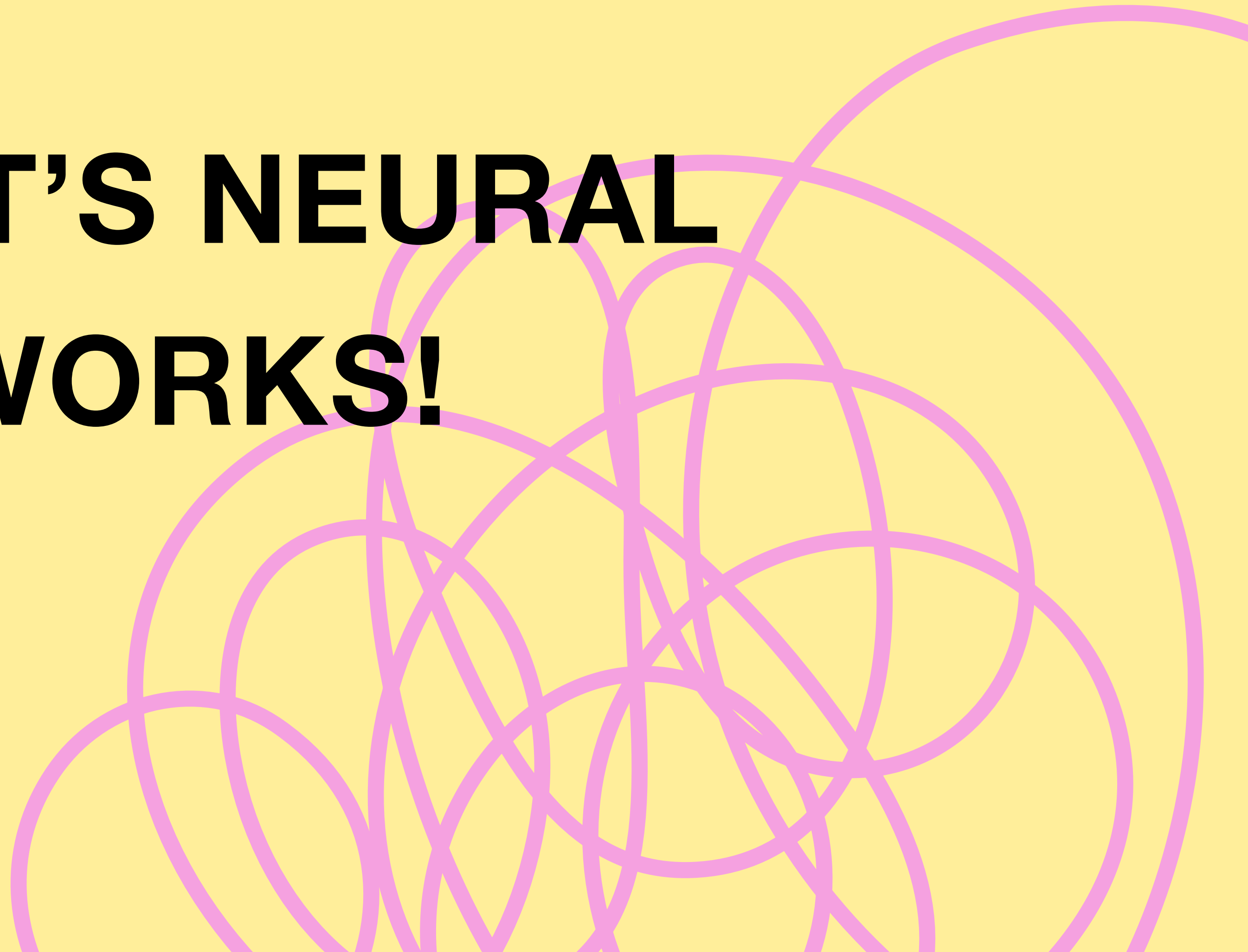
CREDIT ASSIGNMENT PROBLEM

The problem of determining how the performance of an entire system (i.e., the neural network) can be attributed to its individual components (i.e., its parameters).

OUR GRADIENT FROM EARLIER IS HERE TO HELP.



**AND THAT'S NEURAL
NETWORKS!**



SOME FURTHER REMARKS



MAKING THE ALGORITHM EFFICIENT: STOCHASTIC GRADIENT DESCENT.

With large training datasets, it is quite inefficient to compute the gradient for each individual data point and then apply the backpropagation procedure.

Therefore, the dataset is divided into so-called *mini-batches*, i.e., smaller data packages. Gradient descent and backpropagation are then performed for each mini-batch and iterated over all batches.

THAT MUST NEED A LOT OF COMPUTING POWER...

CAN MY COMPUTER EVEN DO THIS?

Yes. Computers are and are becoming increasingly efficient.

Nevertheless, it depends on the specifications....

SOME RECOMMENDATIONS.

An introduction to neural networks on YouTube with wonderful illustrations by 3Blue1Brown:

https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

An interactive visualization of neural networks by Tensorflow (a Python machine learning library):

<https://playground.tensorflow.org/#activation=tanh&batchSize=10&dataset=circle®Dataset=reg-plane&learningRate=0.03®ularizationRate=0&noise=0&networkShape=4,2&seed=0.52441&showTestData=false&discretize=false&percTrainData=50&x=true&y=true&xTimesY=false&xSquared=false&ySquared=false&cosX=false&sinX=false&cosY=false&sinY=false&collectStats=false&problem=classification&initZero=false&hideText=false>

16:00 - 16:30
QUESTIONS.
DISCUSSION.